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SECP3123 - BUSINESS INTELLIGENCE

SECTION 2

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PROJECT PROPOSAL

TITLE:

**BUSINESS INTELLIGENCE ANALYTICS FOR U.S. E-COMMERCE PERFORMANCE
OPTIMIZATION: A MULTI-DIMENSIONAL ANALYSIS OF SALES, CUSTOMER
BEHAVIOUR, AND DELIVERY OPERATIONS (2019–2025)**

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1. Introduction

The U.S. e-commerce industry has experienced rapid growth in recent years. Online sales in the e-commerce industry exceed 1.1 trillion USD by 2024. As online marketplaces continue to expand, businesses must manage complex operations that involve large volumes of customer transactions, product inventories and logistics operations. While these platforms generate massive data every day, turning that data into meaningful insights becomes a major challenge.

Without an effective Business Intelligence (BI) solution, organizations may find it difficult to understand the causes of delivery delays, identify key revenue contributors, and evaluate factors that influence customer satisfaction.

This project aims to address these challenges by transforming raw e-commerce data into valuable business insights. We are using a dataset containing one million U.S. e-commerce orders from 2019 to 2025, the project will analyse sales performance, product profitability, logistics efficiency, and customer purchasing behavior. The findings are aims to support data-driven decision-making by helping businesses improve profitability, optimize logistics processes, and deliver a better overall customer experience.

2. Methodology & Workflow

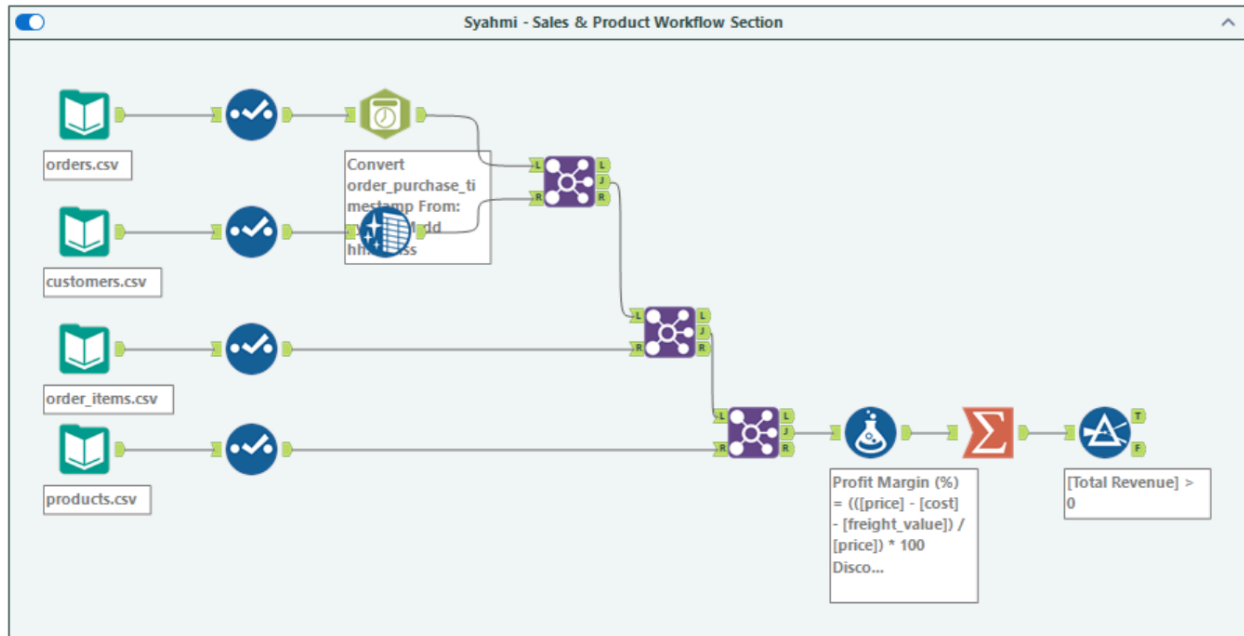
2.1 Data Integration, Cleaning, and Transformation

We used Alteryx Designer to do data processing. We integrate multiple structured CSV files (Orders, Products, Customers, Payments, Sellers, Delivery, and Reviews) into one unified dataset.

- **Data Cleansing:** The Data Cleansing tool was utilized to remove duplicates, null values, and inconsistent text formatting like trailing whitespaces.
- **Data Type Conversion:** Fields like `order_purchase_timestamp` and payment values were initially imported as text and were converted to appropriate `DateTime` and `Double` using the `DateTime` and `Select` tools.
- **Data Integration:** Datasets were merged using `Join` tools linked by `order_id` and `customer_id`.

2.2 Workflow & Tool Configuration

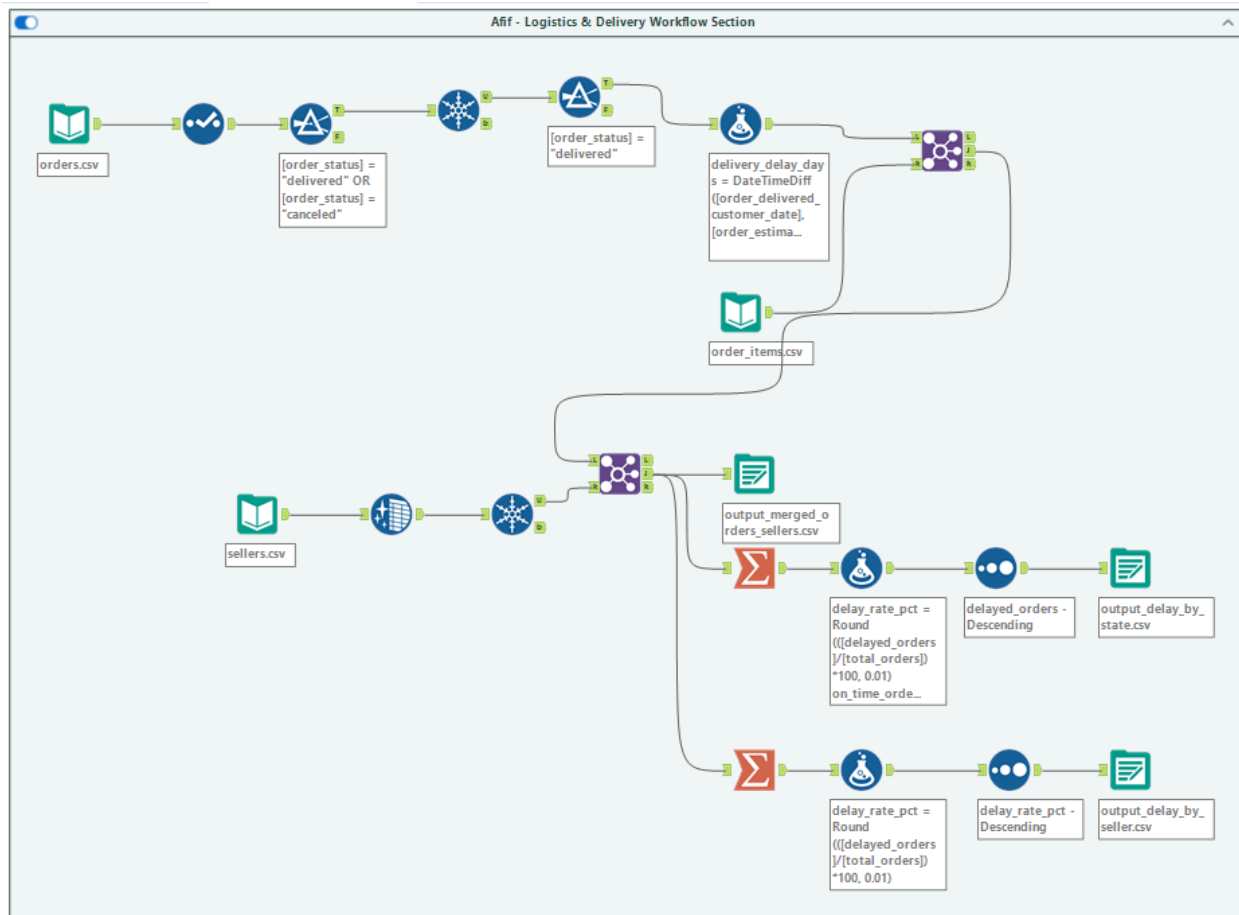
The data processing workflow was developed in Alteryx Designer and divided into three analytical modules: Sales and Product Analytics, Logistics and Delivery Analytics, and Customer and Payment Analytics.



Appendix A: Alteryx Workflow – Sales and Product Analytics

Module 1: Sales and Product Analytics

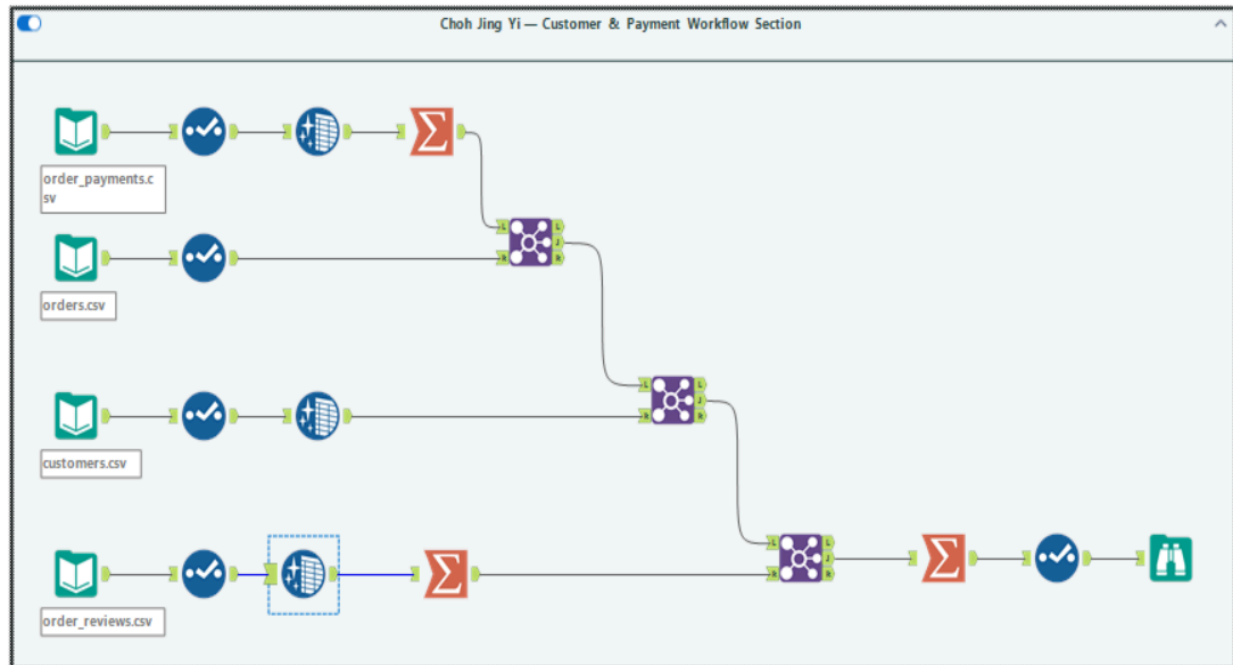
This module focused on analyzing revenue performance and product profitability. The orders, order items, customers, and products datasets were imported using the Input Data tool and cleaned using the Select and Data Cleansing tools. The datasets were then integrated through a series of Join tools using `order_id`, `customer_id`, and `product_id`. A Formula tool was configured to generate derived fields including Profit Margin (%), Discount Rate (%), and Order Year. Finally, the Summarize tool was used to aggregate total revenue by product category, year, and customer state, while the Filter tool removed records containing invalid or zero revenue values.



Appendix B: Alteryx Workflow – Logistics and Delivery Analytics

Module 2: Logistics and Delivery Analytics

The logistics module was designed to evaluate delivery performance and identify operational bottlenecks. Join tools were used to merge the data from the order, sellers and order items datasets. The Filter tool was used to standardize and remove invalid transactions from date-related fields before they were integrated. A Formula tool was created to compute days of delay as the difference between the actual delivery date and the estimated delivery date. Additional calculated fields such as delayed order indicators and delay percentages were created to support performance measurement. The Summarize tool summarized the delivery data by seller and state and the Sort tool ranked regions and sellers by delivery. The processed results were then exported for dashboard development.



Appendix C: Alteryx Workflow – Customer and Payment Analytics

Module 3: Customer and Payment Analytics

This module focused on understanding customer payment behavior and satisfaction levels. The customers, orders, order payments, and order reviews datasets were imported and prepared for integration. The payment records had one to many relationships with orders. The Summarize tool was first used to aggregate the payment values by order_id to avoid data duplication in the join. The data sets were joined with Join tools on the order_id and customer_id. Finally, a Summarize tool was created to provide important key metrics such as total orders, total revenue, and average review scores for each payment method.

2.3 Results

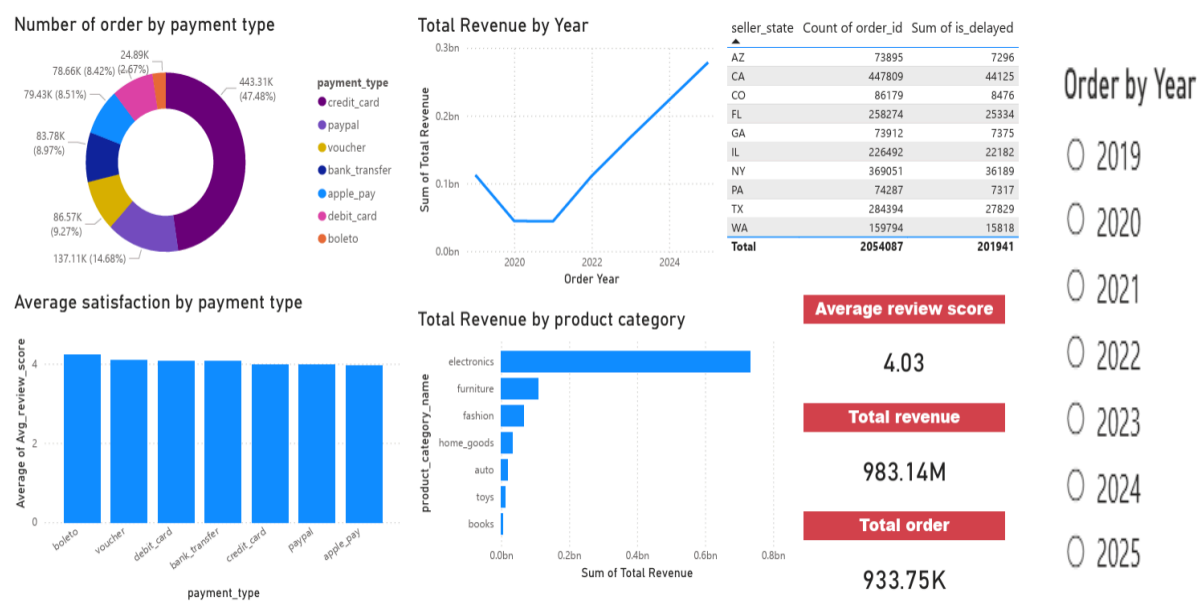
The Alteryx workflows successfully transformed multiple raw e-commerce datasets into structured analytical outputs that were ready for visualization in Power BI dashboards. Through the integration of orders, customers, products, payments, reviews, and seller datasets, The project produced several key performance indicators (KPIs) related to sales performance, customer behavior, payment preferences, and logistics operations. The data preparation process eliminated

duplicate records, corrected data type inconsistencies, and generated calculated fields such as profit margin, discount rate, and delivery delay.

The final outputs were a single dataset on products, payment method performance, customer satisfaction surveying, and delivery performance monitoring. These processed datasets allowed interactive dashboards to be developed, which can be used as tools to help make business decisions. The results provided a comprehensive view of revenue generation, customer purchasing patterns, payment preferences, and delivery efficiency across the e-commerce platform.

3. Visualization & Insights

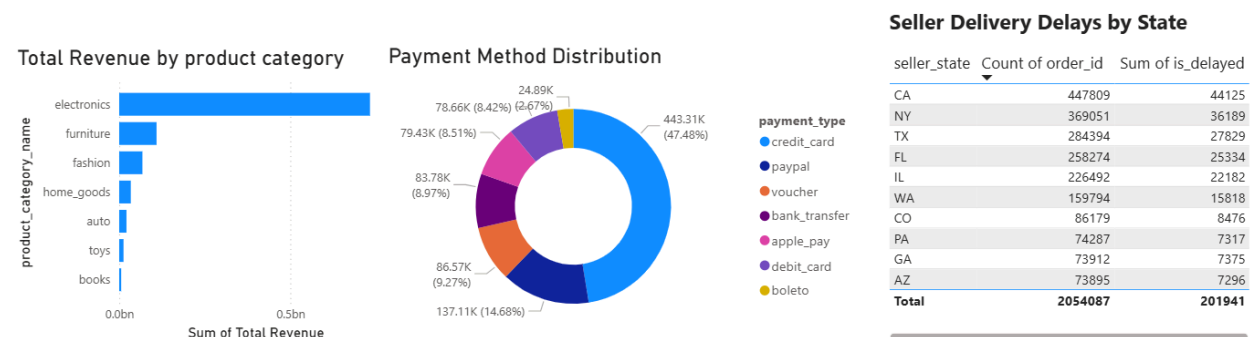
3.1 Dashboard & Storyboard Screenshots



Appendix D: Power BI Dashboards

The dashboards were developed to visualize e-commerce KPIs such as revenue trends, payment distributions and delivery delay by state. We include visualization techniques such as donut charts, bar charts, line charts and tables to provide a clear understanding of customer behavior, payment preferences, product performance, and revenue trends.

The *E-Commerce Customer Journey* storyboard combines the main visualizations to provide a complete overview of business performance. It helps managers understand how revenue trends, customer payment behaviors, and delivery operations are connected, making it easier to identify opportunities for improvement and support strategic planning.



Appendix E: "The E-Commerce Customer Journey" Storyboard

3.2 Charting Choices

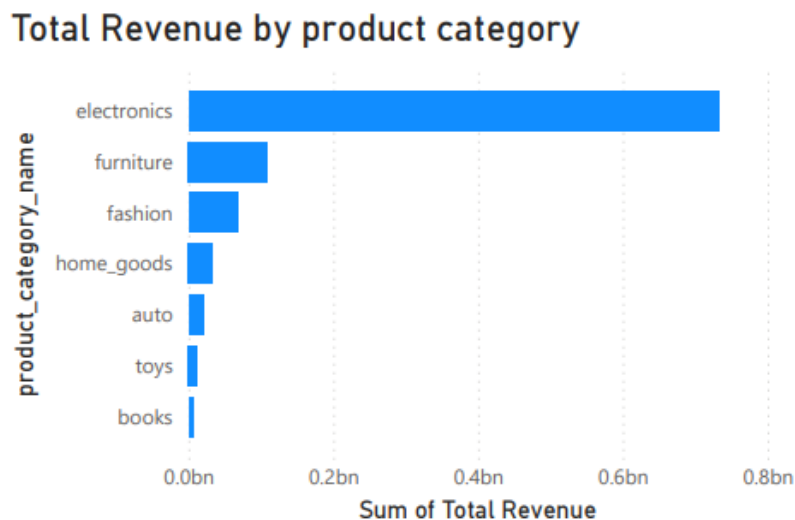
- **Line Charts (revenue trend over time)** were used to present revenue trends over time. This allows stakeholders to clearly observe growth patterns, seasonal fluctuations, and overall business performance from 2019 to 2025.
- **Bar Charts (total revenue by category and average satisfaction by payment type)** were used for categorical comparisons such as total revenue by product category and order volume by state. This makes it easy to identify top-performing categories and high-activity regions.
- **Donut Charts payment method distribution)** were used to display payment method distribution. This helps visualize the proportion of each payment type and identify payment preferences among customers.
- **Tables and KPI Cards** were used to present total revenue, total orders, and average review scores, ensuring that key metrics are easily accessible.
- **Slicers:** Slicers for Order by Year to allow users to interact dynamically with the data. This interactivity enables managers to filter the views and isolate specific regional or temporal performance without needing multiple static reports.

3.3 Visualization Best Practices

- **Clarity and Simplicity:** Each chart was designed to focus on a single insight, avoiding unnecessary complexity or overcrowding of information.
- **Consistent Formatting:** Uniform color themes, labels, and number formatting were used across all dashboards to improve readability and professional presentation.
- **Appropriate Chart Selection:** The most suitable chart type was chosen based on data structure
- **Use of Labels and Tooltips:** Data labels and interactive tooltips were included to provide detailed information without cluttering the visual space.
- **Interactive Elements:** Filters and slicers were added in the dashboard to allow users to explore data by year.

3.4 Insights & Suggested Solutions

3.4.1 Total Revenue by Product Category



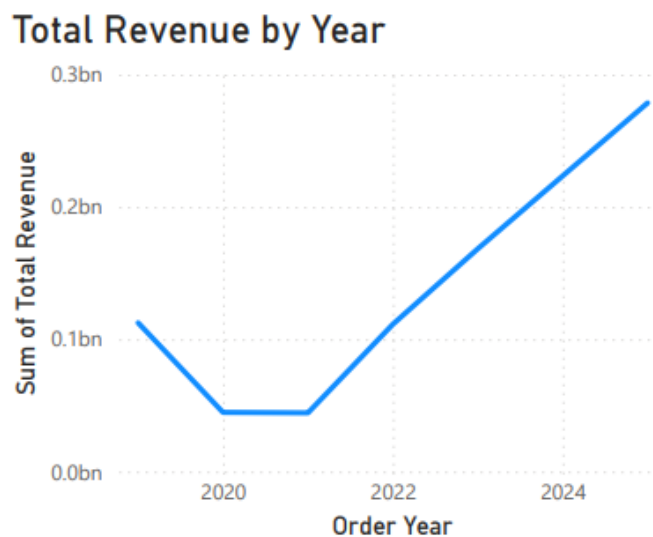
Appendix F: Total Revenue by Product Category Chart

Business Insight: **Electronics** is the dominant revenue category, generating approximately **732.78 million**, far exceeding all other categories. **Furniture** is **109.09 million** and **fashion**

67.82 million rank second and third, while home goods, auto, toys, and books contribute smaller shares. The heavy concentration in electronics presents a revenue risk should demand in that segment decline.

Business Recommendation: The company should strengthen its electronics supply chain to protect its primary revenue stream. Simultaneously, targeted promotions and cross-selling strategies such as bundling electronics with furniture or home goods which can help grow underperforming categories and reduce revenue concentration risk.

3.4.2 Revenue Trend Over Time



Appendix G: Total Revenue by Year Chart

Business Insight: Revenue grew steadily from 2020 through 2022-2023, driven by post pandemic e-commerce expansion. Growth has since begun to plateau, suggesting the business is approaching organic market saturation within its existing footprint.

Business Recommendation: To re-accelerate growth, the company should introduce a customer loyalty programme to improve retention and repeat purchase rates. Geographic expansion beyond the current 10-state footprint and personalised marketing based on existing order data can help unlock new revenue opportunities.

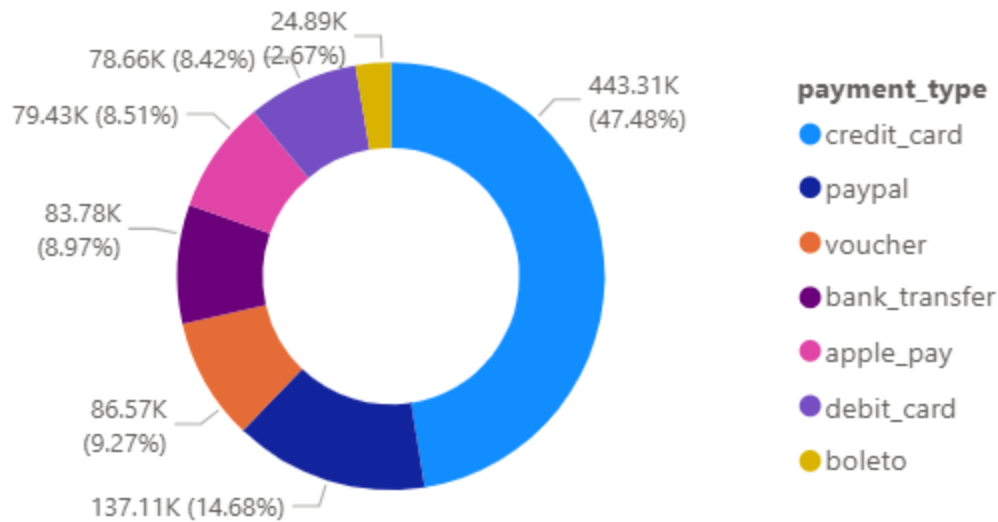
3.4.3 Order Volume and Delivery Delay Rate by State

seller_state	Count of order_id	Sum of is_delayed
AZ	73895	7296
CA	447809	44125
CO	86179	8476
FL	258274	25334
GA	73912	7375
IL	226492	22182
NY	369051	36189
PA	74287	7317
TX	284394	27829
WA	159794	15818
Total	2054087	201941

Business Insight : The regional shows that **California (CA)** handles the highest order volume with **447,809 orders (21.79% of total volume)**, followed by **New York (NY)** with **369,051 orders (17.97%)** and **Texas (TX)** with **284,394 orders (13.85%)**. Together, these three states account for over half of the company's total order volume, confirming their role as the primary operational hubs in the network. These same high-volume states also record the highest absolute number of delayed orders and California alone accounts for **44,125 delayed orders**, followed by New York (**36,189**) and Texas (**27,829**). However, when delay counts are converted into a delay rate (delayed orders ÷ total orders per state), a different pattern emerges: every state falls within a narrow band of **9.79% to 9.98%**, with **Georgia** recording the highest rate at **9.98%** and **Texas and Illinois** the lowest at **9.79%**. Across the entire network, **201,941 of 2,054,087 orders were delayed**, an overall delay rate of **9.83%**. The narrow spread between the highest and lowest state-level delay rates indicates that delivery delays are not concentrated in any particular region, but instead reflect a systemic issue affecting roughly one in every ten orders regardless of where the order is fulfilled.

Business Recommendation: Because California, New York, and Texas drive the majority of order volume, the company should prioritise warehouse capacity, carrier contracts, and staffing in these states to prevent operational strain as volume continues to grow.

3.4.4 Payment Method Distribution



Appendix H: Payment Distribution Chart

Business Insight: The payment method distribution shows that **credit card** is the dominant payment option in nearly half of all transactions about **443.31K orders (47.48%)**. This indicates that customers strongly prefer credit cards due to their convenience, familiarity, and widespread acceptance. **PayPal** ranks as the second most popular payment method with **137.11K orders (14.68%)** highlighting a significant customer segment that prefers digital wallet solutions for faster and more secure online payments. The remaining payment methods include **voucher (9.27%)**, **bank transfer (8.97%)**, **Apple Pay (8.51%)**, and **debit card (8.42%)**, contributing relatively similar shares suggesting that customers value having multiple payment options available. Meanwhile, **boleto** records the lowest usage at **24,889 orders (2.67%)** indicating limited customer preference for this payment method. Overall, the distribution demonstrates that while credit cards remain the primary payment choice, alternative digital payment methods collectively account for a substantial portion of transactions, reflecting diverse customer payment preferences.

Suggestion Solution: As the majority of transactions depend on credit card payments, ensuring the reliability, security, and availability of the credit card payment gateway should be a top priority for the IT and operations teams. Any disruption to this service could have a substantial

impact on revenue generation. In addition, the company may consider promotional initiatives, such as cashback offers or discounts, to encourage customers to use lower cost payment methods like Bank Transfers or Debit Cards, helping to reduce transaction processing expenses.

4. Discussion

The analysis of the e-commerce dataset revealed several important findings related to sales performance, customer behavior, and delivery operations.

4.1 Product Sales Performance

The Electronic category generated the highest revenue among all product categories, making it the company's main source of income. This shows strong customer demand for electronic products. However, relying too much on a single category may create risks if sales in that category decrease in the future. Therefore, the company should also focus on promoting other categories such as Furniture and Fashion to achieve more balanced growth,

4.2 Revenue Growth Trend

The revenue trend shows steady growth from 2020 until 2023, indicating positive business performance. However, the growth rate has started to slow down in recent years. This may suggest that the business is reaching its current market limit. To maintain growth, the company can introduce customer loyalty programmes, personalised promotions, and expand its market reach.

4.3 Delivery and Payment Behaviour

The delivery analysis found that approximately 10% of orders were delivered later than expected. Since the delay rate is similar across all states, the issue may be related to the overall logistics process rather than a specific location. In addition, the payment analysis shows that credit cards are the most preferred payment method among customers. This highlights the importance of maintaining a secure and reliable payment system while continuing to support alternative payment methods.

5. Conclusion

This project successfully applied Business Intelligence techniques to analyse a large U.S e-commerce dataset from 2019 to 2025. Using ALteryx for data preparation and Power BI for visualization, the raw data was transformed into meaningful information that helps explain business performance, customer behavior, payment preferences, and delivery operations.

The findings show that Electronic is the leading revenue contributor, credit cards are the most popular payment method, and delivery delays remain an area that requires attention. The developed dashboard provides a clear overview of key business metrics and supports data-driven decision-making. Overall, this project demonstrates how Business Intelligence tools can help businesses improve performance, identify opportunities, and support future growth.

6. References

Lim, J. (n.d.). *Synthetic U.S. E-Commerce Dataset (2019–2025, 1 Million Orders)*. Kaggle. Retrieved from <https://www.kaggle.com/datasets/limjeongeun/synthetic-u-s-e-commerce-dataset-1m-orders>.

7. Appendix

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